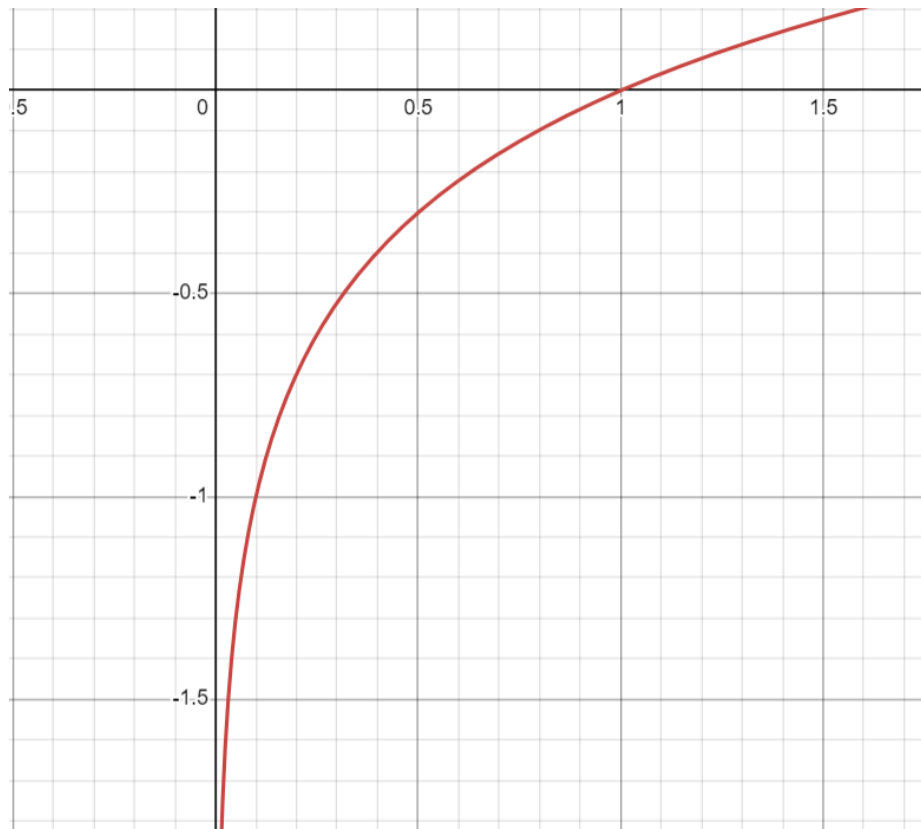


Decision Tree

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Graph of $\log(P(x))$



Graph of $-\log(P(x))$



Information theory

In information theory, chaos processes more information.

Information theory quantifies the amount of information present.

Information of an event is defined as:

$$I(x) = -\log(P(x))$$

Expected Value Vs Average

Toss Fair Coin: 1000 times where 1 represents HEAD
0 represent TAILS

$$\text{Average of all tosses : } \frac{1+0+0+1+1+ \dots + 1}{1000} = \frac{500 * 1 + 500 * 0}{1000}$$

$$= \frac{500 * 1 + 500 * 0}{1000} = \frac{500}{1000} * 1 + \frac{500}{1000} * 0 = \frac{1}{2} * 1 + \frac{1}{2} * 0$$

$$= H * P(H) + T * P(T) = \sum_{x \in \{H,T\}} x * P(x)$$

Expected Value Vs Average

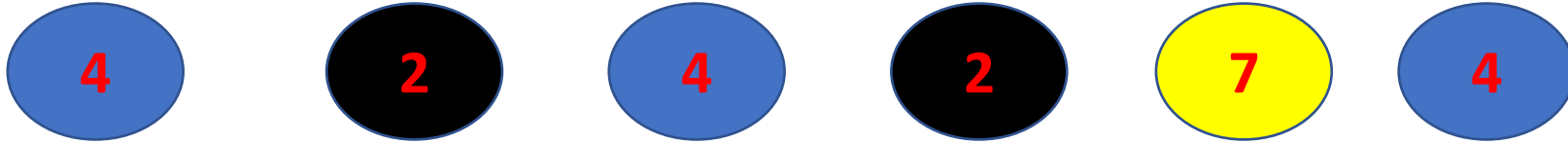
Toss Fair Dice: 6000 times

$$\text{Average of all tosses : } \frac{1+2+1+5+6+ \dots + 4}{6000} = \frac{1000 * 1 + \dots + 1000 * 6}{6000}$$

$$= \frac{1000 * 1 + \dots + 1000 * 6}{6000} = \frac{1000}{6000} * 1 + \dots + \frac{1000}{6000} * 6 = \frac{1}{6} * 1 + \dots + \frac{1}{6} * 6$$

$$= 1 * P(1) + \dots + 6 * P(6) = \sum_{x \in \{1,2,3,4,5,6\}} x * P(x)$$

Expectation of Data Set: Example



$$\begin{aligned}\frac{\sum_i^m x_i}{m} &= \frac{4+2+4+2+7+4}{6} = \frac{1}{6} * 4 + \frac{1}{6} * 2 + \frac{1}{6} * 4 + \frac{1}{6} * 2 + \frac{1}{6} * 7 + \frac{1}{6} * 4 \\ &= \frac{3}{6} * 4 + \frac{2}{6} * 2 + \frac{1}{6} * 7 \\ &= p(4) * 4 + p(2) * 2 + p(7) * 7 \\ &\approx \sum_i p(x_i) * x_i\end{aligned}$$

Entropy: Expected value of Information

In information theory, entropy measures the amount of information.

We define entropy as:

$$H(x) = \sum_x P(x)I(x) = \sum_x P(x)(-\log(P(x)))$$

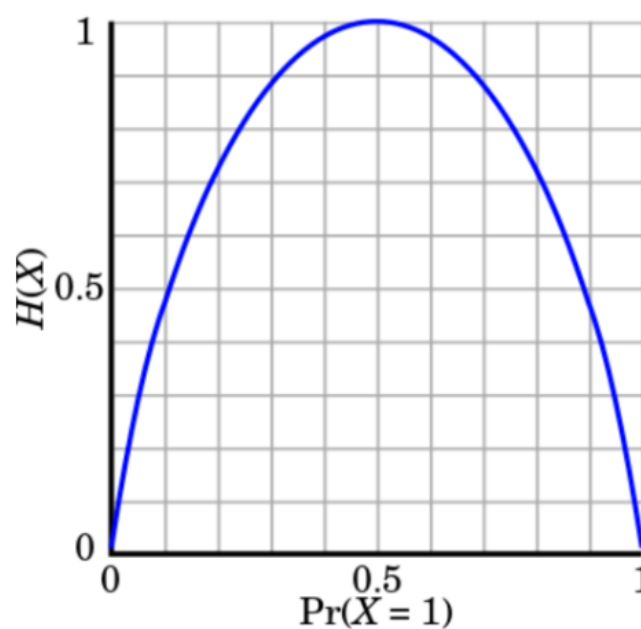
$$H(x) = - \sum_x P(x) \log P(x)$$

Example of Entropy

Let's compute the entropy of a coin. For a fair coin:

$$H(X) = -p(head) \cdot \log_2(p(head)) - p(tail) \cdot \log_2(p(tail)) = -\log_2 \frac{1}{2} = 1$$

The entropy of a coin peaks when $p(head) = p(tail) = 0.5$.

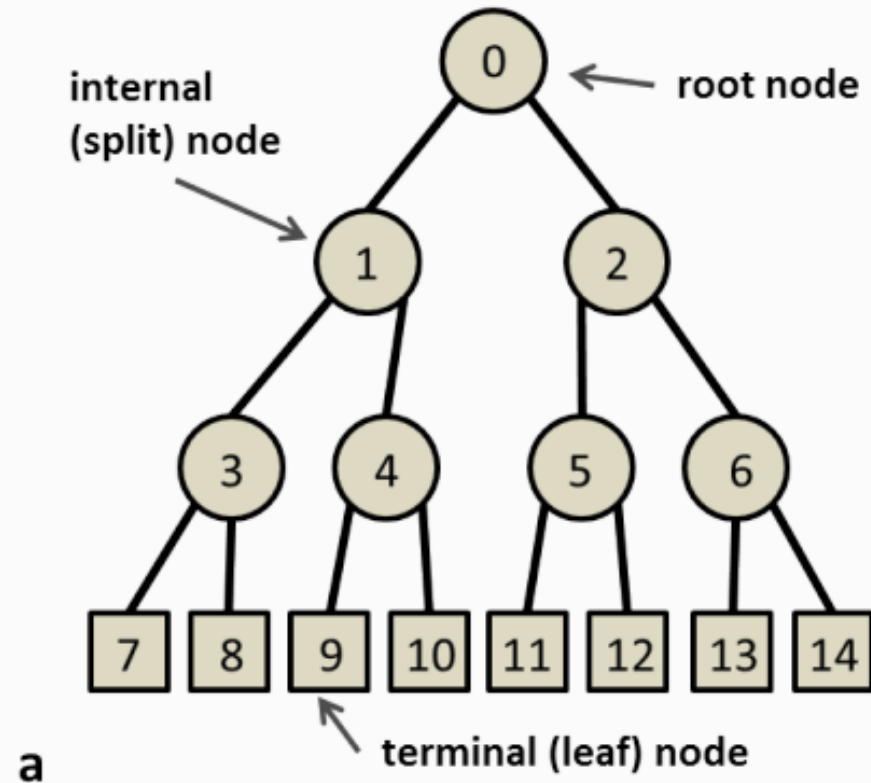


Entropy of Fair Dice?

$$H(X) = \log_2 6 \approx 2.59$$

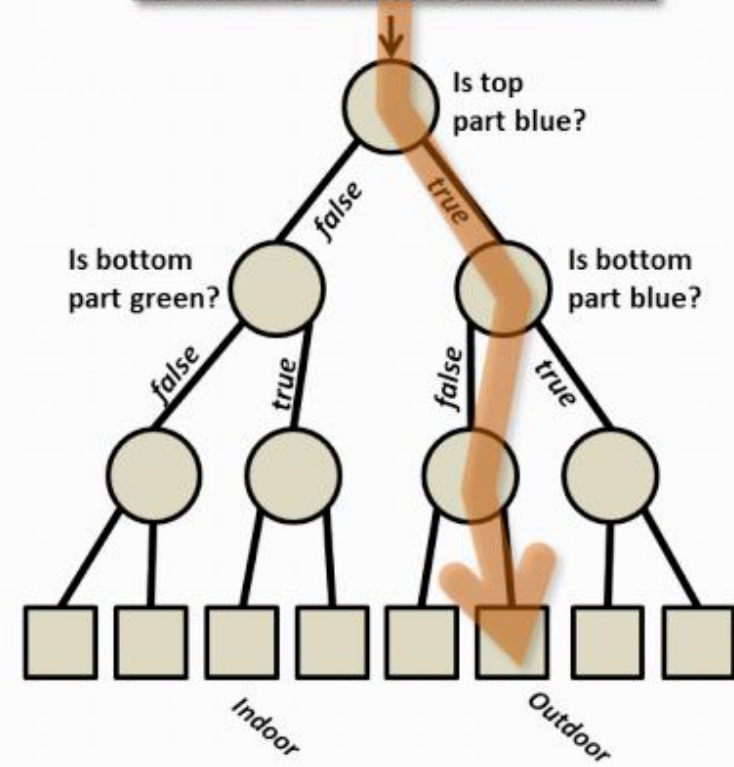
Decision Tree

A general tree structure



Decision Tree

A decision tree



b

Effect of Split based on feature

- Assume 50 Boys and 50 Girls in the class. (Let's represent BOY by 0
And GIRL by 1)

BOYS

Height = 6 feet

Weight: 25 Boys equals to 40 KG
25 Boys equals to 60 KG

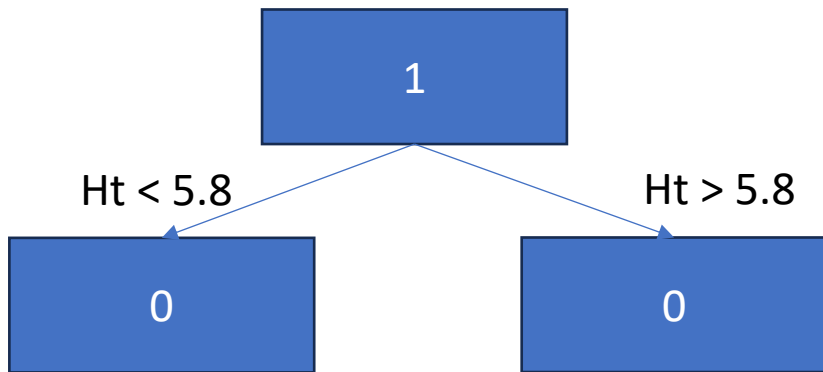
GIRLS

Height = 5.5 feet

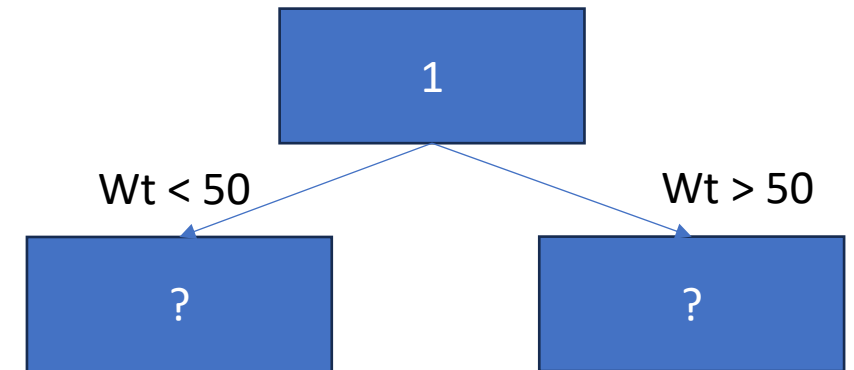
Weight: 25 Girls equals to 40 KG
25 Girls equals to 60 KG

Effect of Split based on feature

$$\text{Entropy of Class [B,G]} = \frac{1}{2} * (-\log \frac{1}{2}) + \frac{1}{2} * (-\log \frac{1}{2}) = 1$$

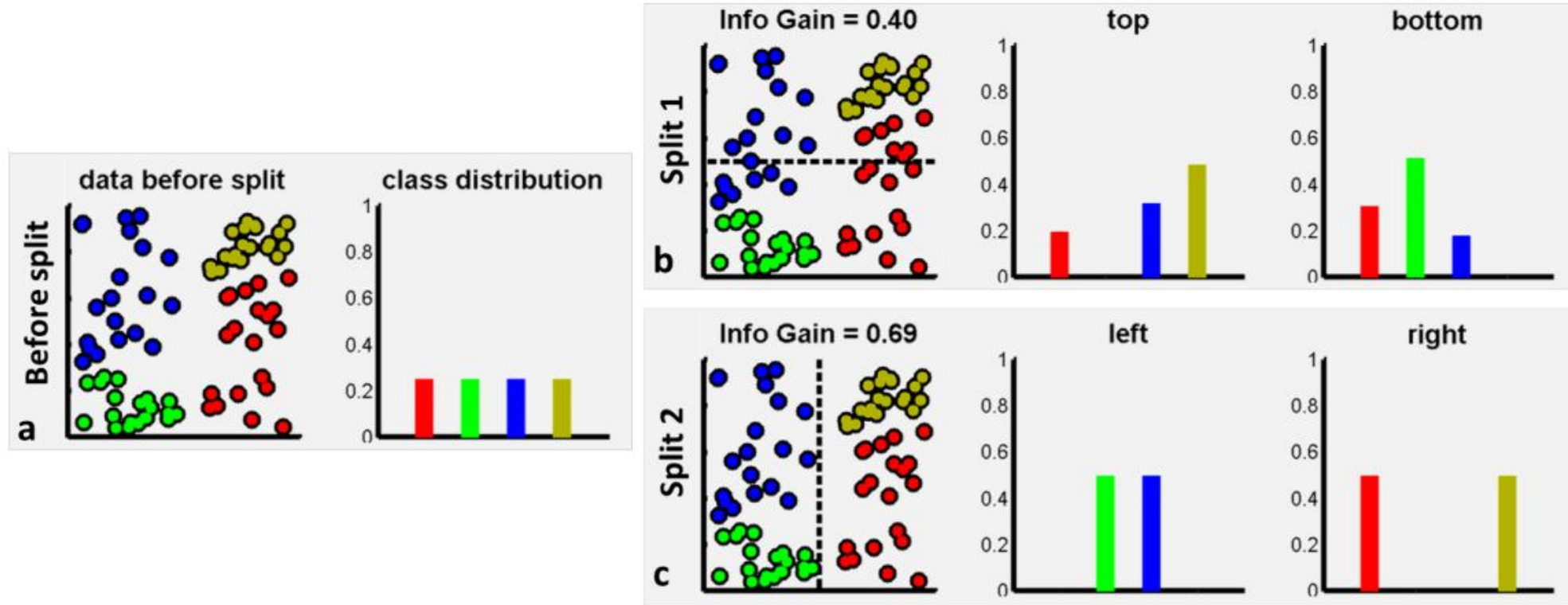


$$\begin{aligned} \text{Total Gain after the Decision} &= 1 - \frac{1}{2} * 0 - \frac{1}{2} * 0 \\ &= 0.5 \end{aligned}$$



$$\begin{aligned} \text{Total Gain after the Decision} &= 1 - ? \\ &=? \end{aligned}$$

Effect of Split based on feature



Information gain for discrete, non-parametric distributions. (a) Dataset $\mathcal{S}$ before a split. (b) After a horizontal split. (c) After a vertical split.

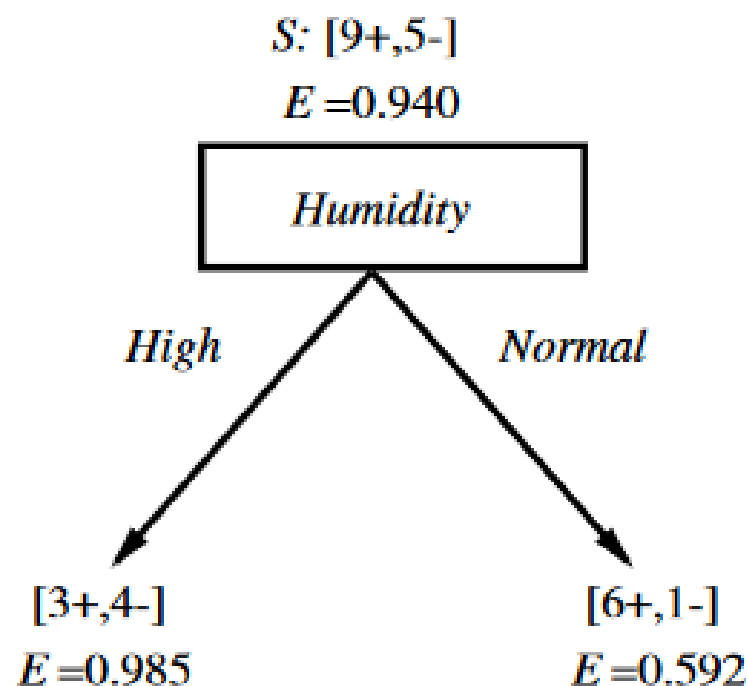
Case study on the following dataset

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

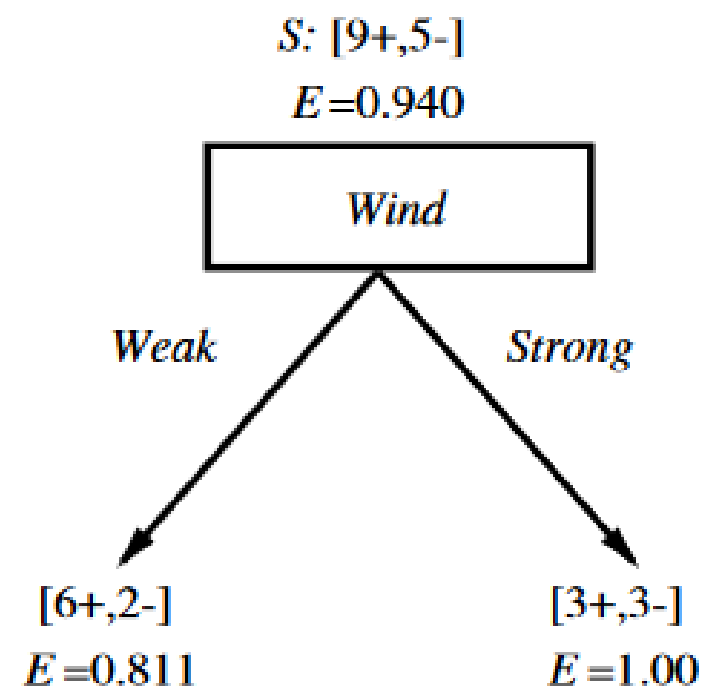
Active

How to choose the best feature?

$$\begin{aligned} \text{Entropy}([9+, 5-]) &= -(9/14) \log_2(9/14) - (5/14) \log_2(5/14) \\ &= 0.940 \end{aligned}$$



$$\begin{aligned} \text{Gain}(S, \text{Humidity}) &= .940 - (7/14).985 - (7/14).592 \\ &= .151 \end{aligned}$$



$$\begin{aligned} \text{Gain}(S, \text{Wind}) &= .940 - (8/14).811 - (6/14)1.0 \\ &= .048 \end{aligned}$$

Calculate the gain for each attribute on the given dataset

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

$$Gain(S, Outlook) = 0.246$$

$$Gain(S, Humidity) = 0.151$$

$$Gain(S, Wind) = 0.048$$

$$Gain(S, Temperature) = 0.029$$

Considering outlook as root node?

Construct the Decision Tree

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

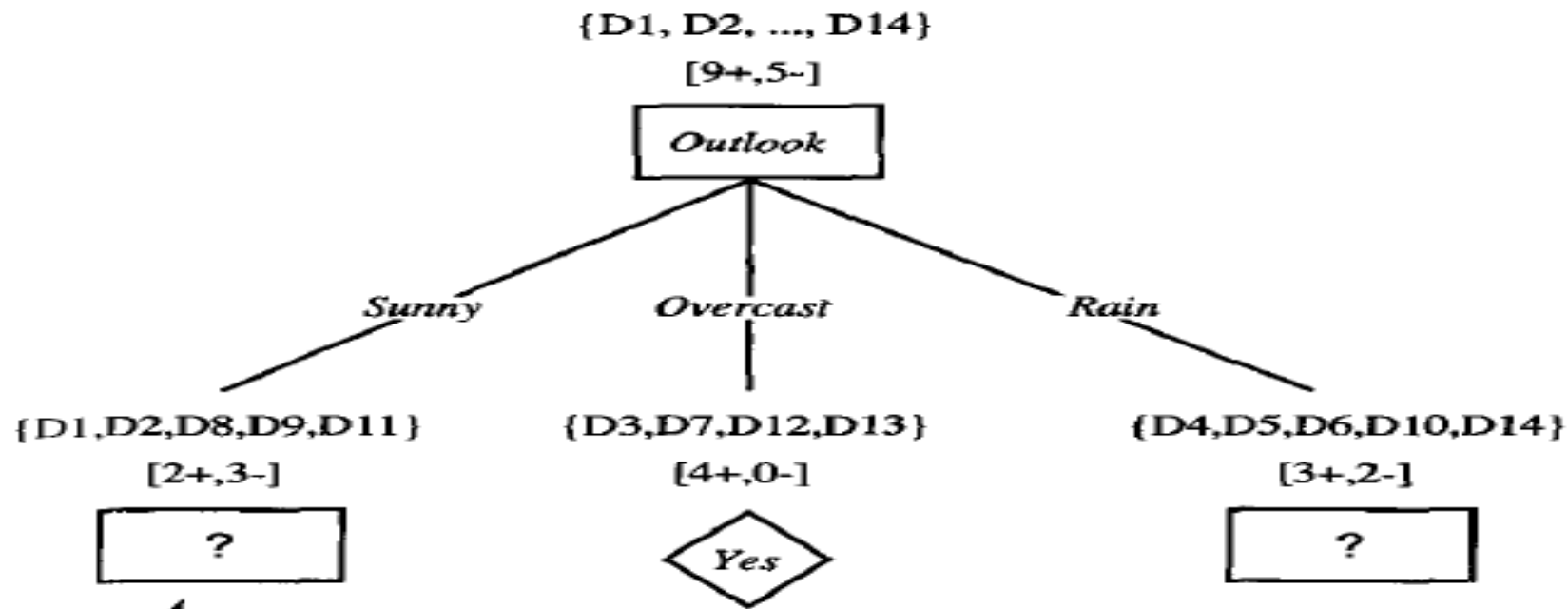
$$Gain(S, Outlook) = 0.246$$

$$Gain(S, Humidity) = 0.151$$

$$Gain(S, Wind) = 0.048$$

$$Gain(S, Temperature) = 0.029$$

How to choose the best feature?



Which attribute should be tested here?

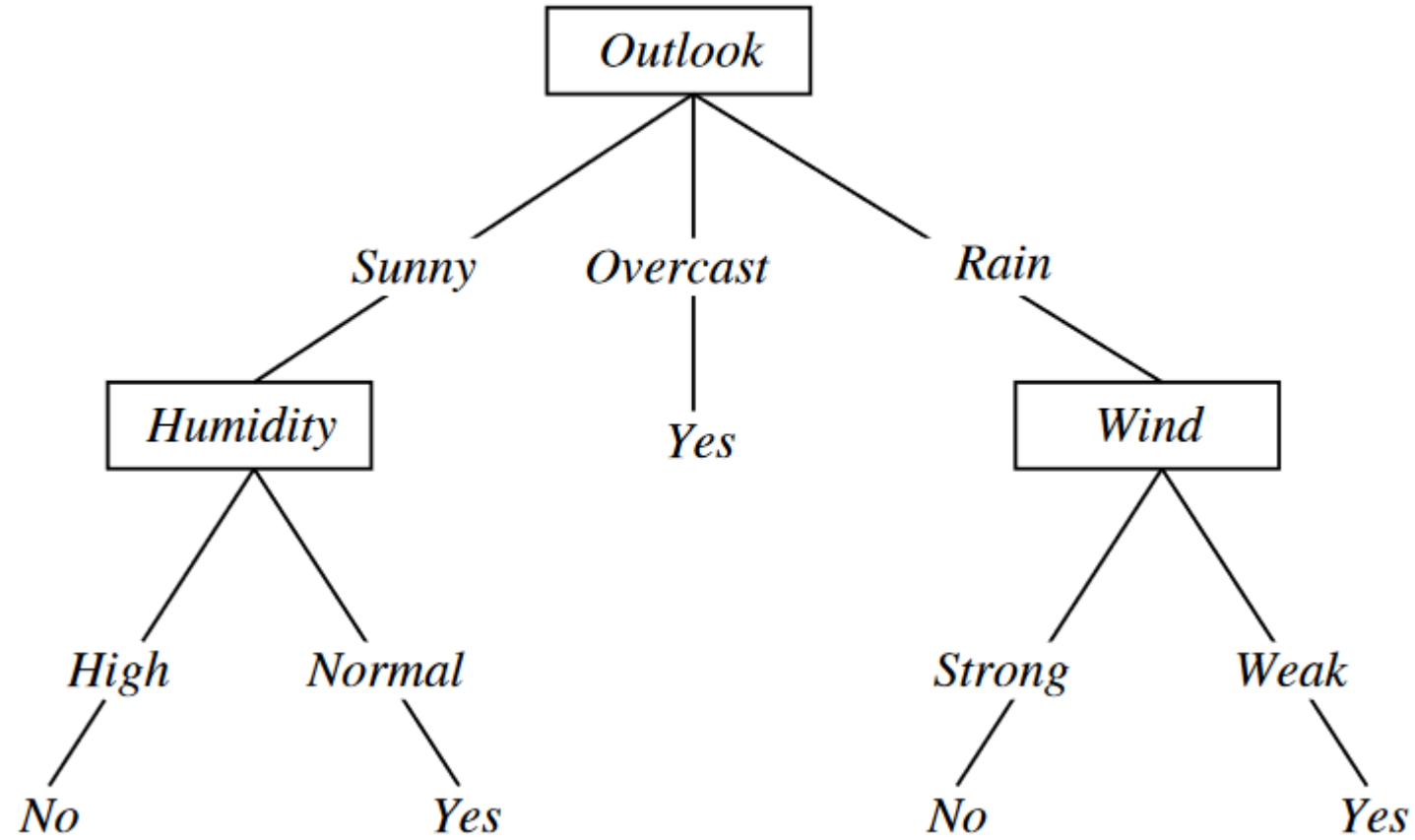
$$S_{\text{sunny}} = \{D1, D2, D8, D9, D11\}$$

$$\text{Gain}(S_{\text{sunny}}, \text{Humidity}) = .970 - (3/5) 0.0 - (2/5) 0.0 = .970$$

$$\text{Gain}(S_{\text{sunny}}, \text{Temperature}) = .970 - (2/5) 0.0 - (2/5) 1.0 - (1/5) 0.0 = .570$$

$$\text{Gain}(S_{\text{sunny}}, \text{Wind}) = .970 - (2/5) 1.0 - (3/5) .918 = .019$$

Decision Tree



ID3(*Examples*, *Target_attribute*, *Attributes*)

Examples are the training examples. *Target_attribute* is the attribute whose value is to be predicted by the tree. *Attributes* is a list of other attributes that may be tested by the learned decision tree. Returns a decision tree that correctly classifies the given *Examples*.

- Create a *Root* node for the tree
- If all *Examples* are positive, Return the single-node tree *Root*, with label = +
- If all *Examples* are negative, Return the single-node tree *Root*, with label = -
- If *Attributes* is empty, Return the single-node tree *Root*, with label = most common value of *Target_attribute* in *Examples*
- Otherwise Begin
 - $A \leftarrow$ the attribute from *Attributes* that best* classifies *Examples*
 - The decision attribute for *Root* $\leftarrow A$
 - For each possible value, v_i , of A ,
 - Add a new tree branch below *Root*, corresponding to the test $A = v_i$
 - Let $Examples_{v_i}$ be the subset of *Examples* that have value v_i for A
 - If $Examples_{v_i}$ is empty
 - Then below this new branch add a leaf node with label = most common value of *Target_attribute* in *Examples*
 - Else below this new branch add the subtree
ID3($Examples_{v_i}$, *Target_attribute*, $Attributes - \{A\}$)
- End
- Return *Root*

Decision Trees

- If X is a vector of n attributes and Y is boolean valued label. How many different functions are possible?
- Overfitting?
 - ✓ Solutions?
- Converting a tree to rules?
- Is hypothesis space is complete?

References

- (Decision Trees): Chapter 3, Tom Mitchell, Machine Learning.

Information Gain Calculations

Play Tennis Dataset – Step-by-step arithmetic
(same style as the textbook diagrams)

1. Entropy of the whole set S

S : [9+, 5-] Total = 14

$$p_+ = 9/14 \approx 0.643 \quad p_- = 5/14 \approx 0.357$$

$$\text{Entropy}(S) = - p_+ \log_2(p_+) - p_- \log_2(p_-)$$

$$= - (9/14) \log_2(9/14) - (5/14) \log_2(5/14)$$

$$= - 0.643 \times (-0.637) - 0.357 \times (-1.485)$$

$$\mathbf{E(S) = 0.940}$$

2. Information Gain of Outlook (Root)

S: [9+,5-]
E = 0.940

Sunny

[2+,3-]
E = 0.971

Overcast

[4+,0-]
E = 0.000

Rain

[3+,2-]
E = 0.971

$$\begin{aligned}\text{Gain}(S, \text{Outlook}) &= E(S) - \sum (|S_v|/|S|) \times E(S_v) \\&= 0.940 - (5/14) \times 0.971 - (4/14) \times 0.000 - (5/14) \times 0.971 \\&= 0.940 - 0.347 - 0 - 0.347 \\&= \mathbf{0.246} \leftarrow \mathbf{highest} \rightarrow \mathbf{choose\ as\ ROOT}\end{aligned}$$

3. Information Gain of Humidity

S: [9+,5-]
E = 0.940

High

[3+,4-]
E = 0.985

Normal

[6+,1-]
E = 0.592

$$\begin{aligned}\text{Gain}(S, \text{Humidity}) &= 0.940 - (7/14) \times 0.985 - (7/14) \times 0.592 \\ &= 0.940 - 0.4925 - 0.296 \\ &= 0.151\end{aligned}$$

4. Information Gain of Wind

S: [9+,5-]
E = 0.940

Weak

[6+,2-]
E = 0.811

Strong

[3+,3-]
E = 1.000

$$\begin{aligned}\text{Gain}(S, \text{Wind}) &= 0.940 - (8/14) \times 0.811 - (6/14) \times 1.000 \\ &= 0.940 - 0.463 - 0.429 \\ &= 0.048\end{aligned}$$

5. Information Gain of Temperature

S: [9+,5-] E = 0.940

Hot

[2+,2-]
E = 1.000

Mild

[4+,2-]
E = 0.918

Cool

[3+,1-]
E = 0.811

$$\text{Gain}(S, \text{Temperature}) = 0.940 - (4/14) \times 1.000 - (6/14) \times 0.918 - (4/14) \times 0.811$$

$$= 0.940 - 0.286 - 0.393 - 0.232$$

$$= 0.029$$

(lowest of the four features)

6. Comparison – Choose the Best Feature

Outlook

IG = 0.246

← MAXIMUM →
ROOT

Humidity

IG = 0.151

Wind

IG = 0.048

Temperature

IG = 0.029

7. After Root = Outlook → Sunny branch

Subset Sunny: [2+, 3-] $E = 0.971$

Humidity (winner)

High → [0+, 3-] $E = 0$

Normal → [2+, 0-] $E = 0$

$$\begin{aligned}\text{Gain} &= 0.971 - (3/5) \times 0 - (2/5) \times 0 \\ &= 0.971\end{aligned}$$

→ Perfect split

High → No

Normal → Yes

Wind (for comparison)

Weak → [1+, 2-] $E \approx 0.918$

Strong → [1+, 1-] $E = 1.000$

$$\begin{aligned}\text{Gain} &= 0.971 - (3/5) \times 0.918 - \\ &\quad (2/5) \times 1 \\ &\approx 0.020\end{aligned}$$

Much smaller → not chosen

8. After Root = Outlook → Rain branch

Subset Rain: [3+, 2-] $E = 0.971$

Wind (winner)

Weak → [3+, 0-] $E = 0$

Strong → [0+, 2-] $E = 0$

$$\begin{aligned}\text{Gain} &= 0.971 - (3/5) \times 0 - (2/5) \times 0 \\ &= 0.971\end{aligned}$$

→ Perfect split

Weak → Yes

Strong → No

Humidity (for comparison)

High → [1+, 1-] $E = 1.000$

Normal → [2+, 1-] $E \approx 0.918$

$$\text{Gain} \approx 0.020$$

Much smaller → not chosen

9. Overcast branch – already pure

Outlook = Overcast

[4+, 0-]

Entropy = 0 (perfectly pure)

No further calculation needed
→ Create leaf that predicts Yes

